

AI-Driven Multidisciplinary Design Optimization BWB Inverse Design

Abstract. A mission profile is mapped directly to a manufacturable blended-wing-body (BWB) design through a data-driven inverse design framework. Since no analytical inverse solution exists, the method combines five forward surrogate models with warm-started differential evolution to optimize 21 variables, including 10 planform and 11 internal structural parameters. Four surrogates are trained from the BlendedNet++ dataset and combined with the provided L/D model. The initial population is seeded using stress-feasible dataset designs, reducing mean loss from 0.4364 to 0.3831 compared with a cold start. Structural feasibility is enforced using a split-conformal stress bound because the stress surrogate under-predicts 44.8% of held-out designs. Volumetric requirements are evaluated using the published loss, while fully compliant alternatives are also reported to quantify the associated mass trade-off. The final pipeline achieves a mean loss of 0.3831 while maintaining the required stress margin for all submitted designs.

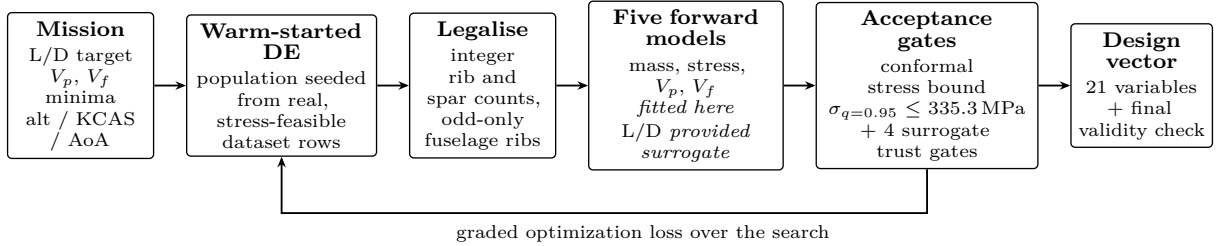


Figure 1: Inverse design pipeline. Mission requirements are passed to a warm-started differential evolution optimizer. Each candidate is first mapped to legal design values and then evaluated using the aerodynamic and structural surrogate models. Candidates are accepted only after satisfying the conformal stress bound and surrogate trust criteria.

1 Optimization strategy

Inputs and outputs. The mission profile specifies the target L/D, minimum payload and fuel volumes, and flight condition (altitude, KCAS, AoA). Payload mass is fixed at 70 kg, with a structural load case of 3 g forward, 2.5 g lateral, and 5 g downward, as defined across all 13,720 dataset samples. The optimizer returns a 21-variable design vector containing 10 planform and 11 structural parameters. Rib and spar counts are integer variables, with fuselage ribs restricted to odd values. Every candidate is mapped to legal values before surrogate evaluation.

Optimization method. The design space is mixed-integer and contains non-smooth feasibility boundaries, particularly through the stress constraint. Since the surrogate models evaluate populations efficiently in batch, differential evolution (DE) was used. The final configuration uses `best1bin`, $NP = 252$, $F \in [0.4, 1.0]$, and $CR = 0.85$, with 200,000 evaluations across seeds 0, 1, and 2.

Warm-start initialization. Instead of initializing DE with a Sobol sample over the full 21-dimensional space, the initial population is drawn from real dataset designs that are stress-feasible and biased toward lower mass. This places the search close to physically meaningful regions and reduces the number of evaluations spent locating feasibility. Under the same surrogate models and evaluation budget, warm-started DE reduced mean loss from 0.4364 to 0.3831.

2 Constraint handling

Alternative	Result	Decision
Uniform random	mean loss 0.5781	rejected
DE, cold start	mean loss 0.4364	rejected
DE, warm start	11/18; loss 0.3831	adopted

Table 1: Solver comparison using the same surrogate models and evaluation budget. Warm-started DE was retained because it achieved low loss without requiring a separate generative model.

2.1 Stress constraint with conformal calibration

The allowable stress is 335.3 MPa for Al 7075-T6 using a safety factor of 1.5, with the implementation using a conservative 335.0 MPa threshold. Stress is the weakest surrogate, with $R^2 = 0.746$ on the group-held-out split and under-prediction on 44.8% of held-out designs. Directly constraining the predicted stress can therefore admit designs whose true stress exceeds the allowable.

Split-conformal calibration is applied to

$$\log_{10} \sigma_{\text{true}} - \log_{10} \hat{\sigma}$$

using held-out planforms. The resulting multiplicative margins are $1.49\times$, $2.48\times$, and $3.65\times$ for $q = 0.75$, 0.90 , and 0.95 , respectively. The final pipeline uses $q = 0.95$, corresponding to an effective predicted-stress limit of approximately 91.9 MPa.

The conformal bound is enforced as an acceptance condition rather than only as a penalty. When the raw stress prediction alone is used for acceptance, optimized designs can appear feasible while their calibrated upper stress reaches 1,136 MPa. The strict conformal gate keeps the submitted designs below the allowable limit.

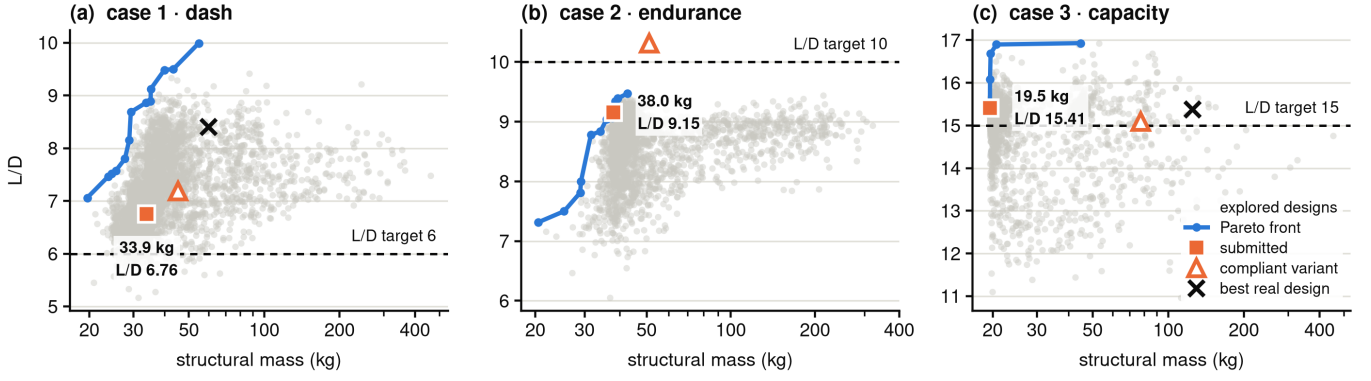


Figure 2: Mass-aerodynamic-efficiency trade-off. The non-dominated designs obtained during optimization are shown for each mission case. Mass and L/D represent only two components of the full objective, so designs outside the projected front may provide greater payload or fuel volume. The compliant variants satisfy all mission targets and remain lighter than the lightest corresponding dataset designs. No dataset design reaches $L/D = 10$ for Case 2 at the required flight condition; the compliant optimized design reaches 10.31.

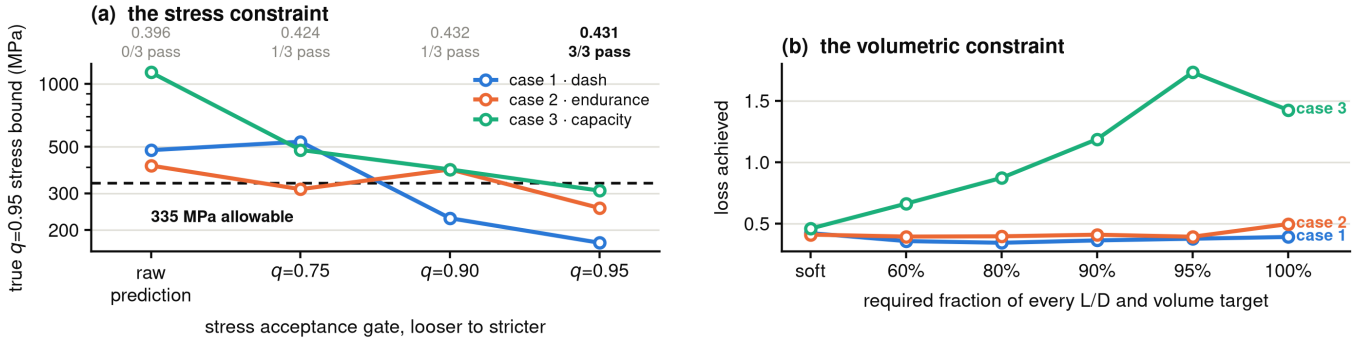


Figure 3: Effect of constraint handling. (a) Stress acceptance using different uncertainty margins. Less conservative gates produce lower optimization losses but may admit designs that exceed the allowable stress after calibration. (b) Loss obtained when L/D and volume requirements are enforced to at least a fraction τ of their targets. Cases 1 and 2 show limited sensitivity, whereas Case 3 exhibits a stronger mass-volume trade-off. These sweeps use 20,000 evaluations with seed 0; the final runs use 200,000 evaluations across three seeds.

2.2 Volume: graded according to the objective

The published loss caps each L/D and volume shortfall at 0.2 while the mass term remains unbounded. Meeting every volume target is therefore not always the loss-minimizing solution when additional structure produces a large mass penalty. The optimizer minimizes the published loss directly.

Figure 3(b) evaluates an additional requirement that each target reach at least a fraction τ of its requested value. Cases 1 and 2 show little change in loss, whereas Case 3 exhibits a stronger and non-monotonic mass-volume trade-off. Fully compliant alternatives are therefore reported alongside the loss-optimal solutions. Their masses are 45.3, 51.1, and 77.7 kg, compared with 59.8, 63.6, and 125.0 kg for the lightest dataset designs satisfying the same missions.

2.3 Surrogate trust constraints

Four additional gates restrict optimization to regions where the surrogates remain reliable: an empirical mass frontier, K-fold ensemble disagreement below 10%, support-based novelty screening, and bound-pinning detection. Without these checks, ensemble disagreement can reach 74%, compared with approximately 1.8% on dataset designs. The gates increase the reported optimization loss but reduce exploitation of poorly supported regions of the surrogate space.

3 Multidisciplinary coupling

Aerodynamic and structural design cannot be treated independently because several variables affect both disciplines. The planform controls L/D while also influencing mass, payload and fuel volume, and structural stress. The centreline chord C_1 , for example, affects aerodynamic performance through Reynolds number and also sets the physical scale of the aircraft. Reducing it may lower mass but can simultaneously reduce L/D and available internal volume.

The internal structure does not directly affect the aerodynamic model, but it controls mass, stress, and usable volume. Increasing structural material generally lowers stress at the cost of additional mass, while removing material can shift the critical stress location. Stress is therefore not assumed to vary monotonically with thickness, whereas eight physically supported monotonic constraints are applied to the mass surrogate.

Each 21-variable candidate is evaluated by all five forward models in the same optimization step so that these trade-offs are resolved simultaneously.

Validation by planform group. Each planform appears two to three times with different internal structures but at the same flight condition. A random split can therefore place closely related geometries in both training and test sets. Group-wise splitting by planform provides a stricter test on unseen geometry and associated flight

conditions. The held-out R^2 values are 0.9889 for mass, 0.9977 for payload volume, 0.9936 for fuel volume, and 0.7460 for stress. Across several model classes, stress prediction remains near $R^2 = 0.72$ – 0.76 , with a near-duplicate analysis suggesting a practical ceiling close to $R^2 = 0.80$. This motivates the use of conformal calibration rather than relying only on the point estimate.

4 Results

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<i>submitted – minimum published loss</i>			
mass (kg)	33.9	38.0	19.5
L/D ach./target	6.76/6	9.15/10	15.41/15
V_p ach./min	0.698/0.75	0.755/0.80	0.321/1.00
V_f ach./min	0.316/0.45	0.367/0.45	0.184/0.65
σ raw / $q = 0.95$	61/234	63/281	89/317
loss	0.3450	0.3690	0.4352
<i>compliant variant – all targets met</i>			
mass (kg)	45.3	51.1	77.7
L/D achieved	7.19	10.31	15.09
V_p / V_f	0.750/0.451	0.909/0.451	1.002/0.655
σ raw / $q = 0.95$	134/267	170/341 [†]	166/329
loss	0.3626	0.4090	0.6217

Table 2: Loss-optimal and fully compliant designs for the three mission cases. Volumes are in m^3 and stresses in MPa, with an allowable stress of 335.3 MPa. [†]The Case 2 compliant design exceeds the $q = 0.95$ conformal bound by 1.8%, although its raw predicted stress remains below the allowable. All reported design vectors were checked for parameter bounds and discrete structural constraints.

The submitted designs achieve losses of 0.3450, 0.3690, and 0.4352 for Cases 1–3, respectively. The objective is dominated by the mass contribution, while the largest volume shortfalls occur in Case 3. The submitted designs remain below the allowable stress after conformal calibration.

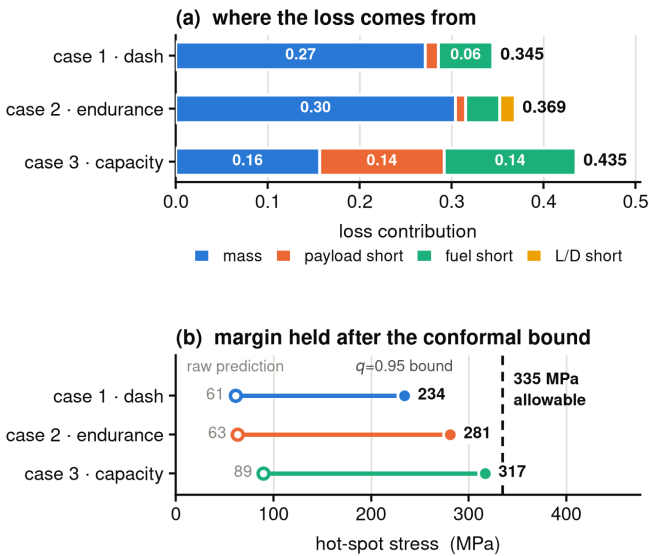


Figure 4: Submitted design performance. (a) Contribution of each objective term to the final loss. (b) Raw stress predictions and their $q = 0.95$ conformal upper bounds relative to the allowable stress. The conformal bound is used for acceptance.

5 Reproducibility

The complete pipeline is implemented in Python and runs on CPU without external CFD or FEA software. After installing the dependencies with `pip install -r requirements.txt`, `run_final.py` maps mission requirements to candidate designs and verifies variable bounds, integer counts, odd-only fuselage ribs, and spar ordering. Hidden mission files can be evaluated without modifying the pipeline. Additional scripts reproduce the constraint studies, solver comparisons, and figures reported here.